



H₂Float

Loadcell Based Floating Pipe Flood Detection

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Develop a Sensor with Flood Detection Capabilities

Continuous Water-Level Sensor Based on Load Cell and Floating Pipe

research done by

Sheng-Wei Wang; Chen-Chia Chen; Chieh-Ming Wu; Chun-Ming Huang

- Cost optimized
- Low power
- Remote monitoring
- Environmentally friendly

Archimedes Principle

$$F_B = \rho_w F_g V$$

$$V = \pi r^2 \cdot d$$

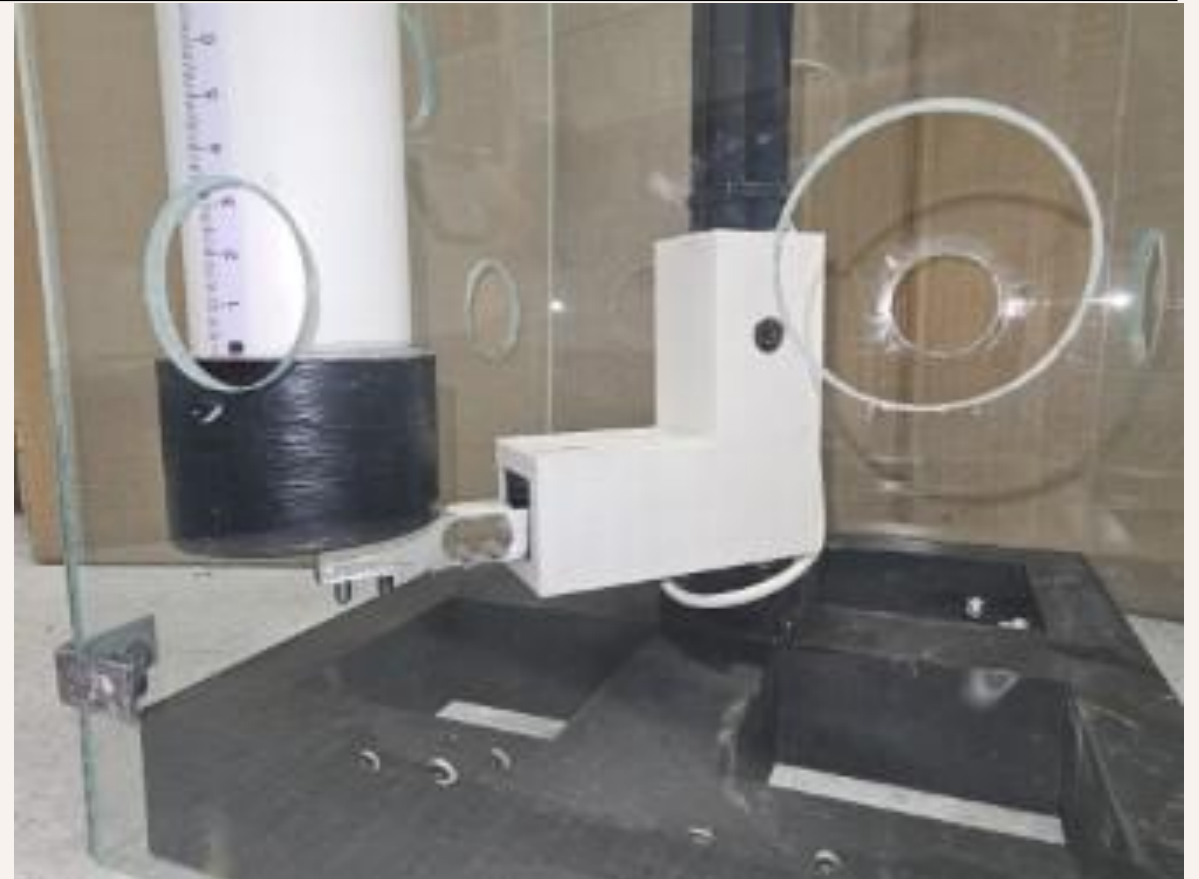
F_B = Buoyant Force

ρ_w = Density of water

F_g = Gravitational Force

V = Volume of displaced water

d = Distance of displacement

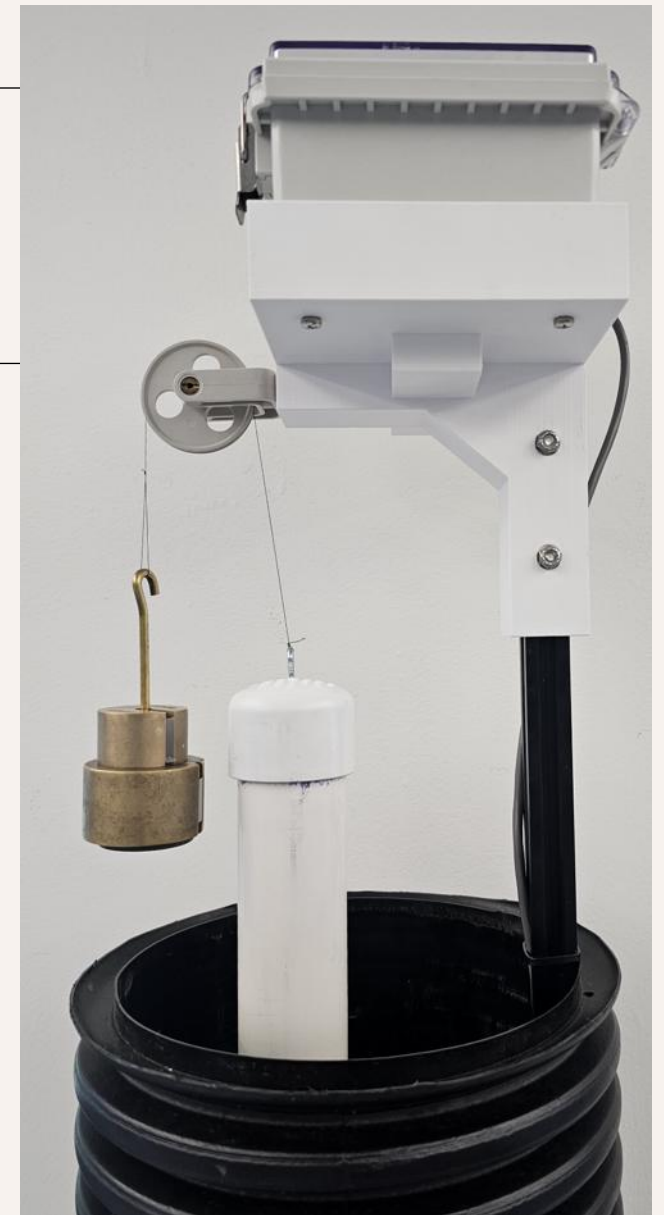


Calibration

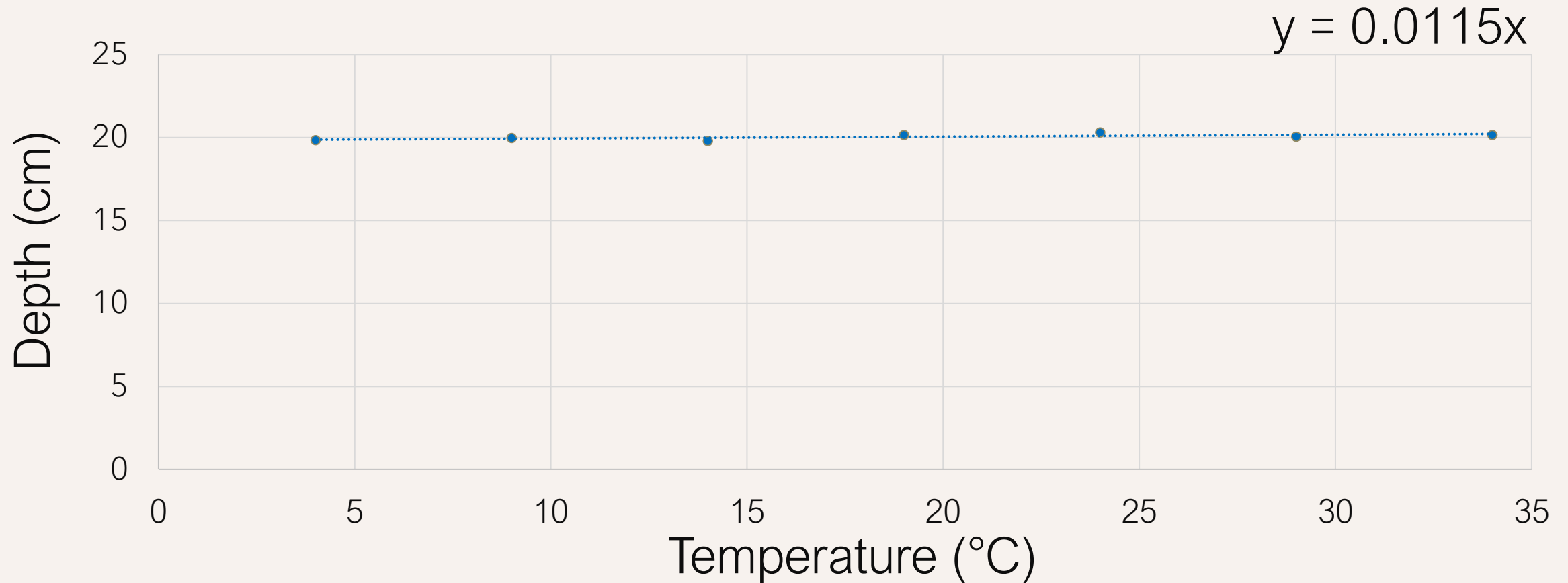
2-point Calibration

- Reading at zero weight (Tare)
- Reading at 750g

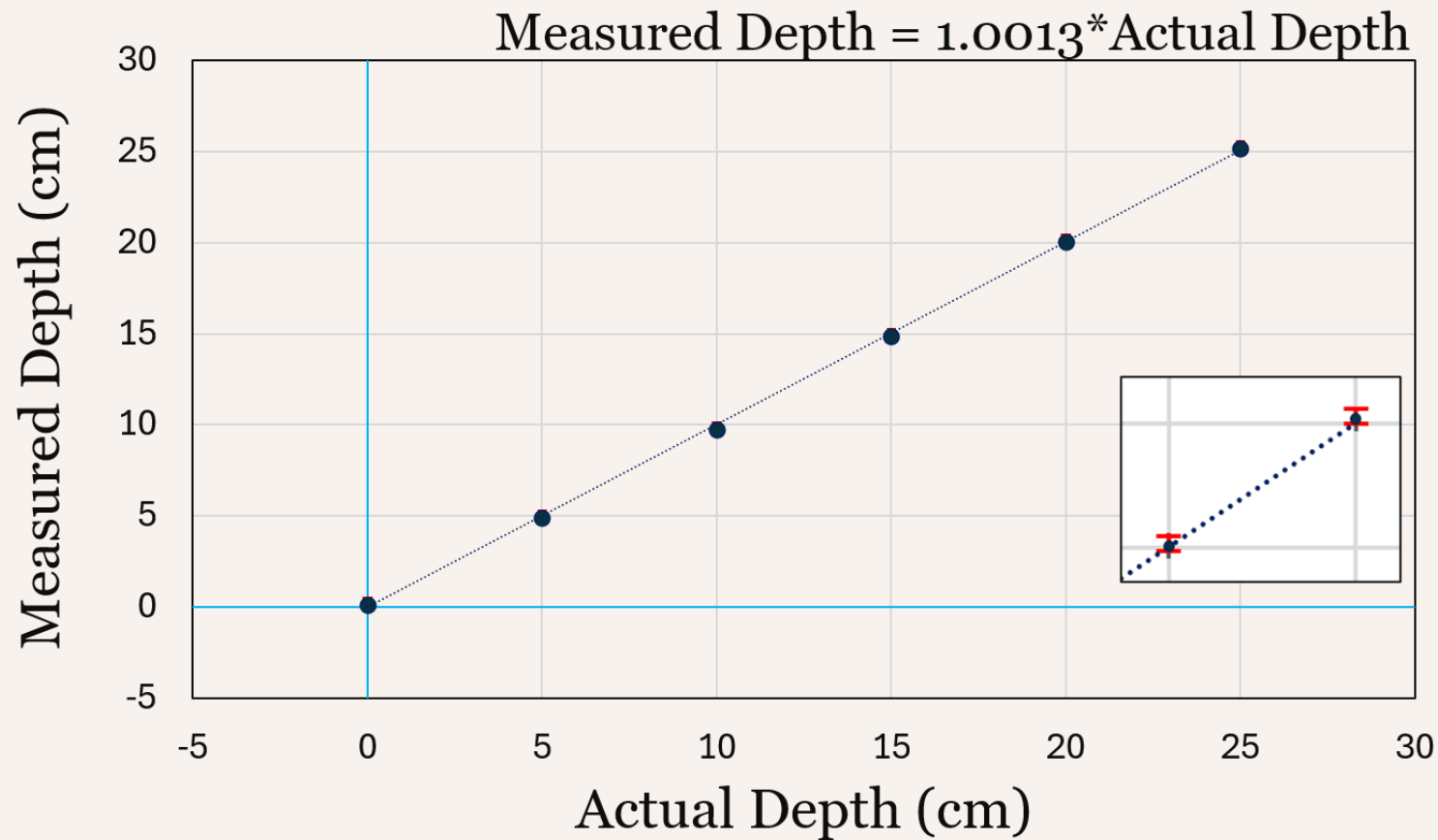
Device is calibrated once during deployment setup



Measured Depth vs Water Temperature



Device Measured Depth vs Tape Measured Depth



Device Features

- LoRa Data sending via gateway
- RTC
- Solar Powered
- Battery monitoring
- Low power



Smart City Implementation

- Bridge Pilings
- Culverts
- Can be programmed to make public safety alerts to first responders and residents of flood conditions based on the data transmitted

Sincere Appreciation

Augusta Utilities and Mr. Wes Byne
for Summer 2024 Internship opportunity

My Mentor Dr. Hauger
Support of AU² Team

Resources

S. -W. Wang, C. -C. Chen, C. -M. Wu and C. -M. Huang, "A continuous water-level sensor based on load cell and floating pipe," 2018 IEEE International Conference on Applied System Invention (ICASI), Chiba, Japan, 2018, pp. 151-154, doi: 10.1109/ICASI.2018.8394554.